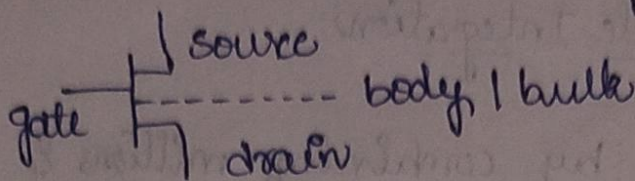
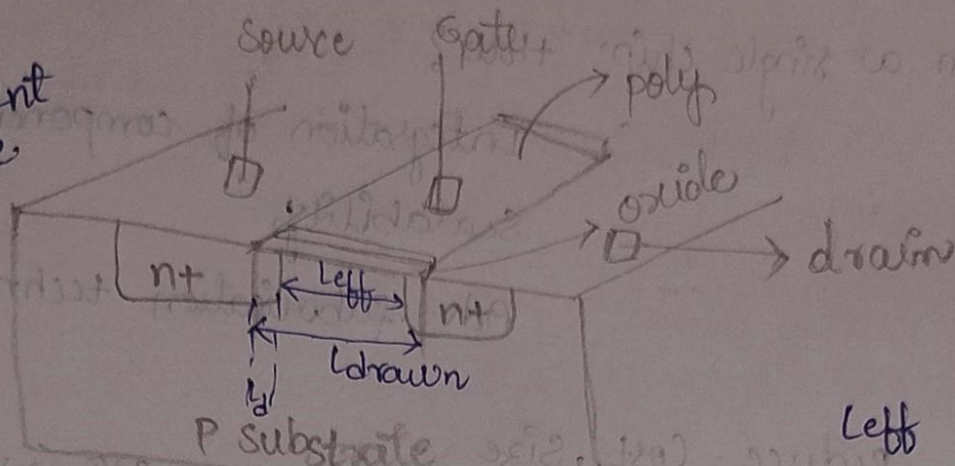


biasing - amt of external source given.



Enhancement type.



$$L_{\text{eff}} = L_{\text{drawn}} - 2L_d$$

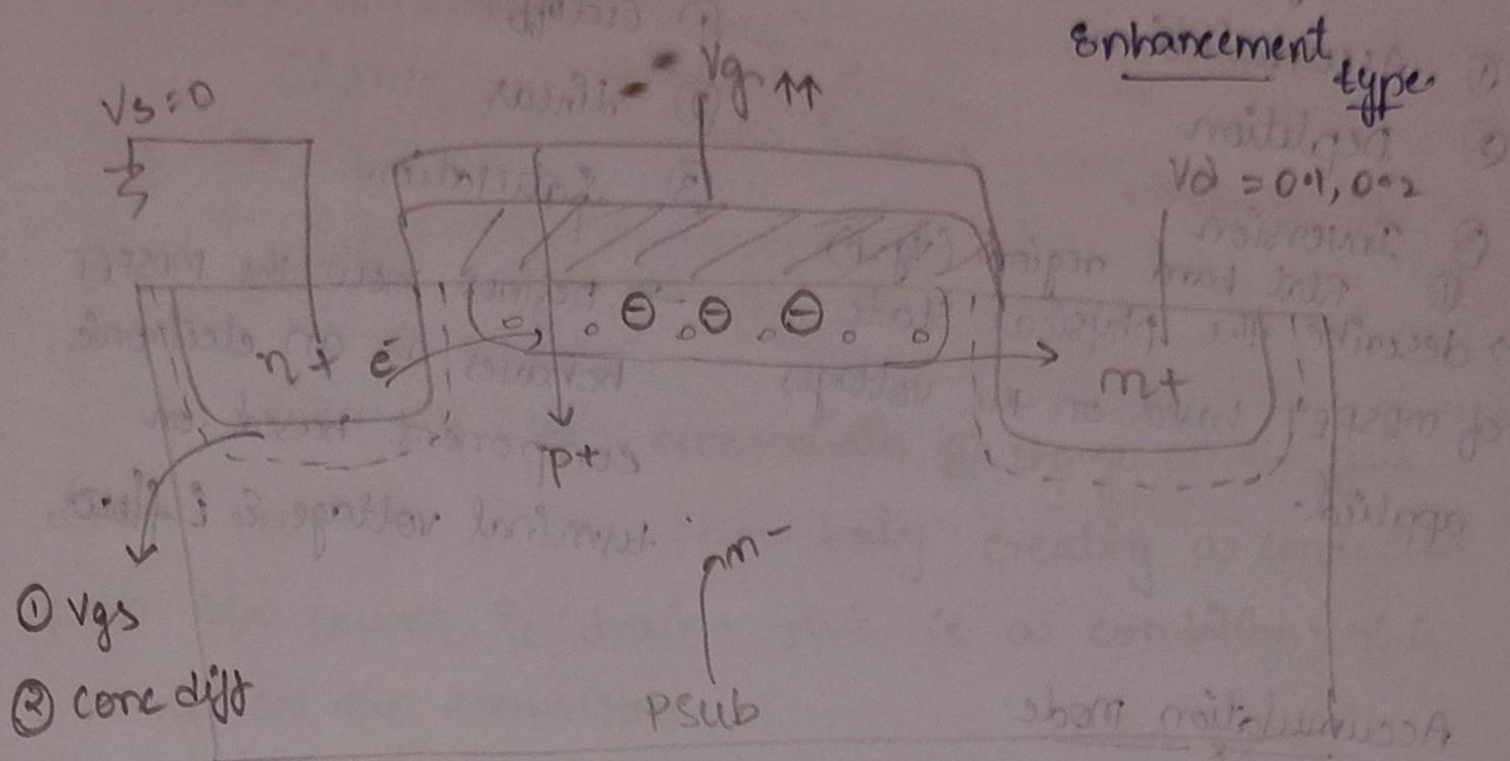
L_d - side diffusion length

In high voltage devices often req the drain to have lighter doping or larger spacing from the gate to handle higher voltages. so source at left, drain at right.

t_{ox} = oxide thickness.

we use n^+ because n^+ represents heavy doping, more no of donor atoms where n^- represents lightly doped with less " donor atoms."

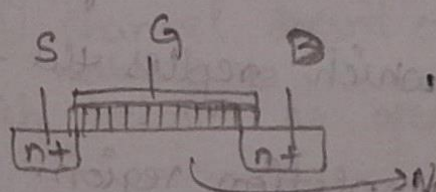
Modes of Operations of MOSFET



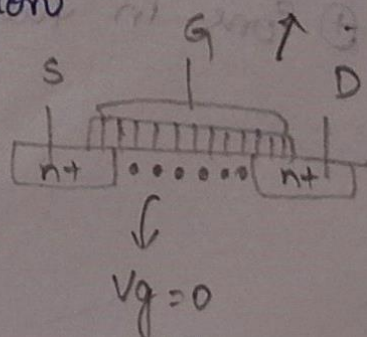
Initially when no V_g, V_d is applied, there forms a depletion region around n^+ with equal distances. which makes no e^- to flow from source to drain terminal making the MOS to be in cutoff region, no current. Later as we increase V_g value the $+$ ve ions below the oxide layer gets attracted forming n^+ ions. so this forms a channel beneath the oxide & where the e^- in source comes out experiencing some potential on increase in V_d the e^- starts moving producing current

NMOS - $V_d, V_g = +ve$ to get current
 body = P-type, source & drain = n^+ region

PMOS - $V_d, V_g = -ve$ to get current
 body = n-type



Source, drain = p^+ region
 $V_g = 0$



Operating modes of MOS.

Regions.

① Accumulation

② Depletion

③ Inversion

④ Flat band region ($V_g = 0$)

→ describe the physical state of MOSFET based on the voltages applied.

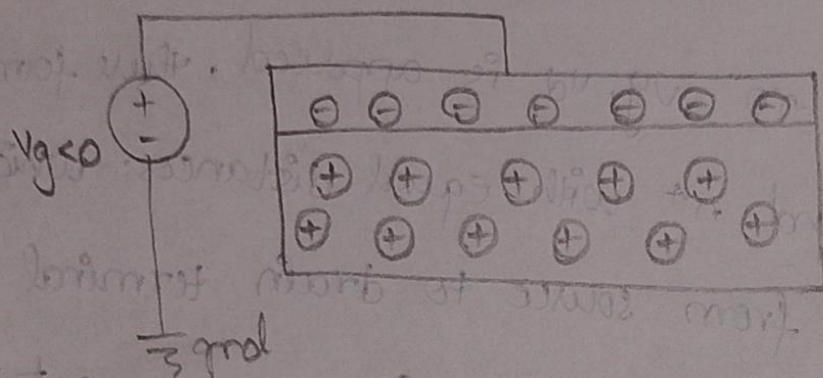
① Cutoff

② Linear (ohmic)

③ Saturation

→ defines how the MOSFET behaves as an electronic component based on terminal voltage & i flow

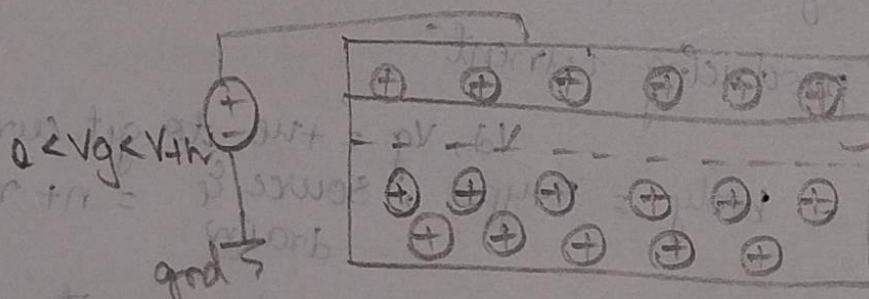
Accumulation mode.



NMOS - off

When $V_g < 0$ for NMOS it's off, accumulates $-ve$ ions on the body which attracts free $+$ ions in the substrate forming accumulation mode MOSFET

Depletion mode.

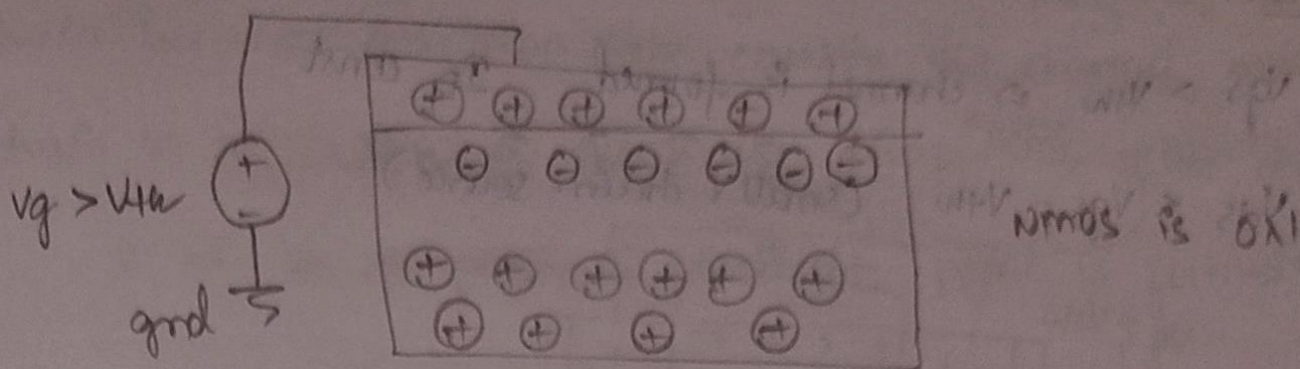


depletion region.

NMOS - off but with little $+$ ve charge on gate

When $V_g < V_{th}$; $0 < V_g$ some small amt of $+$ ve charge is accumulated on the gate which repels the free $+$ ions in the body creating a depletion region.

Inversion mode



When $V_g > V_{th}$ accumulates $\oplus$ charge on the gate which attracts $\ominus$ ions from body creating a conductive path b/w source & drain. This is a condition, this turns ON the transistor.

• Difference b/w Inversion mode & channel.

Inversion mode - phy condition that creates the channel

Enables the creation of a channel.

channel - physical region that facilitates flow.

Serves as pathway for current flow.

Operation Regions.

Cutoff Region

IV characteristics

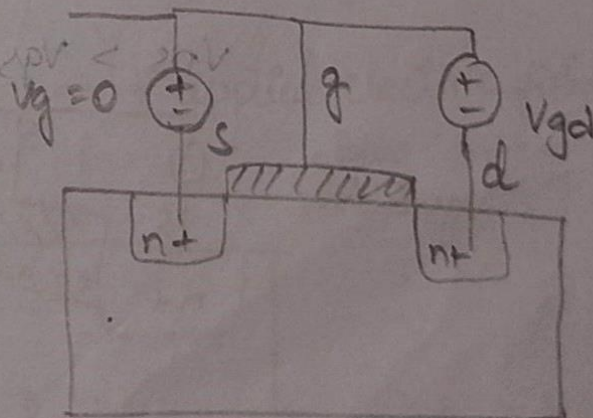
[Long channel] / Ideal MOS

$$V_{gs} < V_{th}, I_D = 0, \text{MOS} = \text{OFF}$$

① V_{gate} is insufficient to attract minority charge carriers to form inversion layer.

So,
② Channel doesn't exist b/w S & D which behaves as open switch

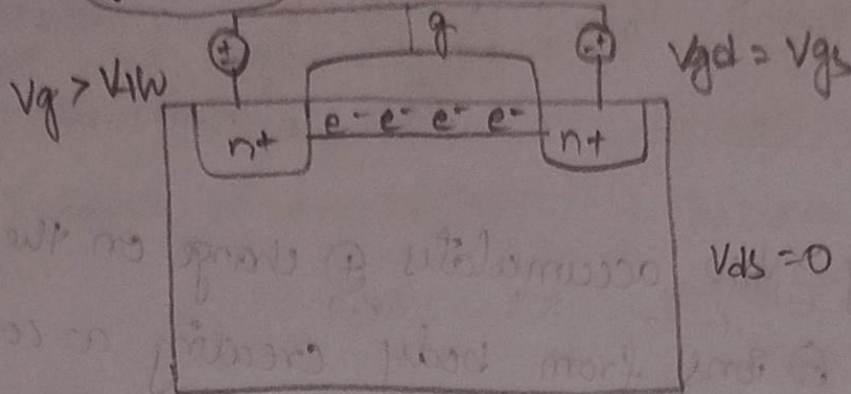
③ Depletion layer prevent by remaining wide.



Linear / Ohmic / Triode

$V_{gs} > V_{th}$ $\Rightarrow$ channel is formed I_D const

$V_{ds} < V_{gs} - V_{th}$ (small V drain-source)



A conductive channel forms in NMOS when $V_{gs} > V_{th}$, due to minority carrier accumulation at gate. Initially this is due to e^- attracted from substrate. During V_{ds} applied conduction, extra e^- come from heavily doped n^+ regions of source & drain.

① $V_{gs} > V_{th}$, a conductive channel forms due to minority carriers at gate, connects source & drain.

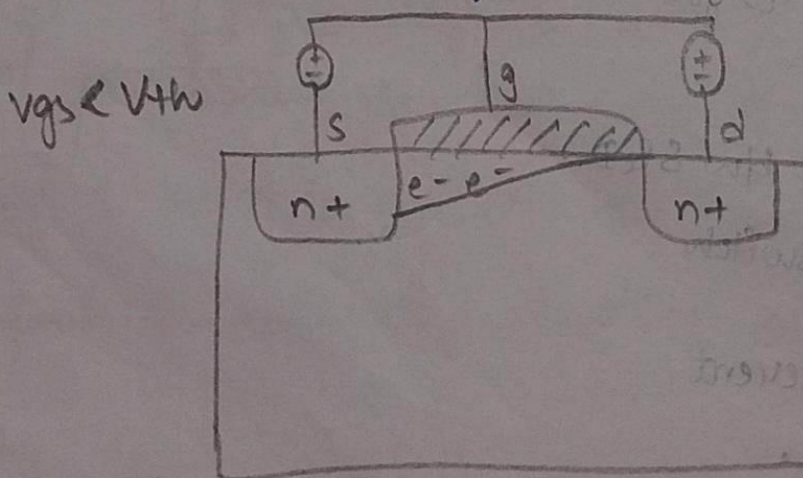
② By applying V_{ds} , it drives e^- through channel enabling i flow b/w source & drain.

③ In linear V_{ds} is small so $I_D \propto V_{ds}$.

Saturation

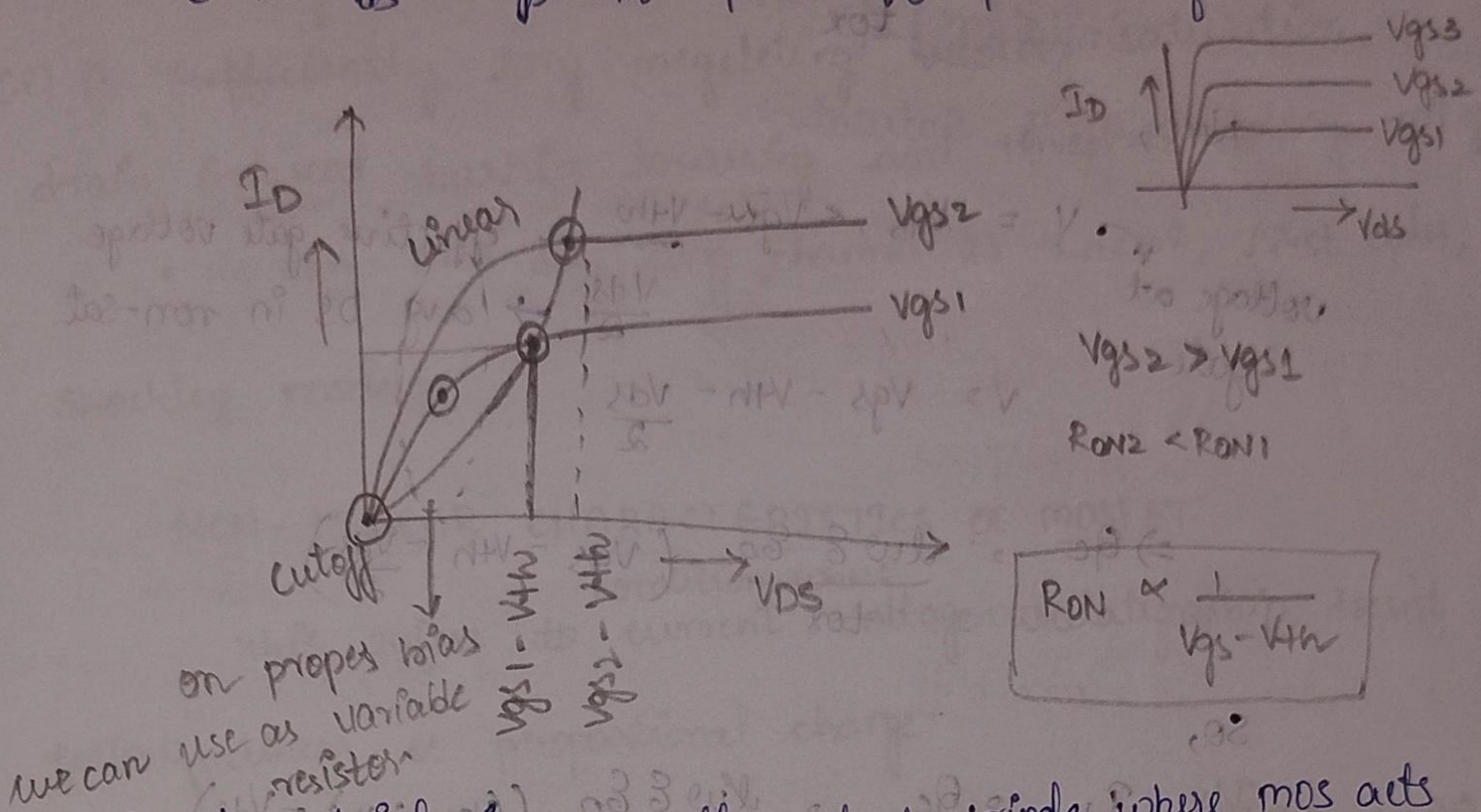
$V_{gs} > V_{th}$

$V_{ds} > V_{gs} - V_{th}$



As V_{DS} increases beyond $V_{GS} - V_{th}$, the voltage diff b/w gate & drain becomes less than V_{th} causing the channel near the drain to pinch off which eventually disappears near drain.

① At $V_{DS} = V_{GS} - V_{th}$ pinch off point is formed.



$$R_{ON} \propto \frac{1}{V_{GS} - V_{th}}$$

The deep triode is a small part of triode where mos acts as near-ideal resistor.

Derive I_D using V_{DS} & V_{GS} .

[Long channel MOSFET]

$$v_{el} = \frac{d \cdot v}{t_{ym}}$$

$$I_{DS} = \frac{\text{charge induced in channel } (Q_c)}{\text{electron transmission time } (T_{sd})}$$

$$T_{sd} = \frac{\text{length } (l)}{v_{e^-}}$$

where $v_{e^-} = \mu E_{ds}$ $\rightarrow$ applied electric field b/w $d \rightarrow s$

$$E_{ds} = \frac{V_{DS}}{l}$$

$$\Rightarrow T_{sd} = \frac{l}{\mu E_{ds}} = \frac{l}{\mu \left(\frac{V_{DS}}{l} \right)} = \frac{l^2}{\mu V_{DS}}$$

where $Q_c = \frac{A \epsilon}{d} \times V$

$\Rightarrow Q = C V$
 $C = \frac{A \epsilon}{d}$

Now,

$Q_c = \frac{\mu \epsilon \epsilon_0}{t_{ox}} \times V$

Now,

$V = V_{ds} < V_{gs} - V_{th}$ effective gate voltage
 $\frac{V_{ds}}{2} \rightarrow$ avg DS in non-sat
 $V = V_{gs} - V_{th} - \frac{V_{ds}}{2}$

$\Rightarrow Q_c = \frac{\mu \epsilon \epsilon_0}{t_{ox}} \left(V_{gs} - V_{th} - \frac{V_{ds}}{2} \right)$

So,

$R = \frac{V}{I} \quad I_{ds} = \frac{Q_c}{t_{ox}} = \frac{\mu \epsilon \epsilon_0}{t_{ox}} \left(V_{gs} - V_{th} - \frac{V_{ds}}{2} \right)$

Linear $\rightarrow$ Resistor (ohmic)

saturation $\rightarrow$ Current source

$I_{ds} = \frac{\epsilon \epsilon_0 \omega}{d l} \mu \left[(V_{gs} - V_{th}) - \frac{V_{ds}}{2} \right] V_{ds}$
 $C_{ox} = \frac{\epsilon \epsilon_0 \omega}{d l}$

$I_{ds} = C_{ox} \frac{\mu}{l} \left[(V_{gs} - V_{th}) - \frac{V_{ds}}{2} \right] V_{ds}$

$I_{ds} = C_{ox} \mu / l \left[V_{ds} (V_{gs} - V_{th}) - \frac{V_{ds}^2}{2} \right] \rightarrow$ Linear

in sat $V_{ds} = V_{gs} - V_{th}$

$I_{ds} = C_{ox} \mu / l \left[\frac{(V_{gs} - V_{th})^2}{2} \right] \rightarrow$ saturation

if $V_{ds} \ll 2(V_{gs} - V_{th})$

$\Rightarrow R_{on} = \frac{1}{\mu} \omega / l C_{ox} (V_{gs} - V_{th})$

Charge sheet model assumption is where we consider the thickness of the inversion region is so small, assuming to be a thin layer.

- A long channel MOSFET is a MOSFET where channel length (L) is sufficiently long neglecting velocity saturation, drain induced barrier lowering and threshold voltage roll off.
- we can also say long channel as Ideal, first order, Shockley model.

NON-LINEAR CHARACTERISTICS OF MOSFET

condition where this refers to, current, voltage relationship doesn't follow linear or proportional change.

① Linear Region - In this region MOSFET acts as variable resistor, but I_D doesn't increase linearly with V_{DS} for all its values.

I_D depends on both V_{DS} & V_{GS} [non-linearity]

② Saturation - In this I_D is independent of V_{DS} .

$(V_{GS} - V_{th})^2$ gives inherent non-linearity.

③ MOSFET → non-ideal
 → non-linear

short long
 - non linear non-ideal, non linear are same.

- non-ideal → Linear, Saturation

→ Linear, Saturation regions.

NON-IDEAL CHARACTERISTICS OF MOSFET (Short channel)

Short channel suffers to reduced size of the channel which raises few limitations / effects.

① Velocity Saturation

In linear region of operation as V_{ds} increases,

$$V_{ds} \uparrow \quad E_x \uparrow \quad V_d \uparrow \quad J \uparrow \quad I_d \uparrow$$

$\hookrightarrow$ electric field $\rightarrow$ current density
 $\hookrightarrow$ velocity of drift.

where,

drift velocity - external
diffusion - conc diff

$$V_d = \mu_n E \quad J = \mu_n E n_q$$

At high electric field, carriers experience scattering due to lattice vibrations which prevents carriers from further acceleration causing their velocity to saturate at

[max]

this reduces the I_D compared to ideal sat region.

② Mobility Degradation.

As channel length is reduced, electric field impacts the mobility of e^- 's. This leads to reduced I_D at high V_{gs} values.

③ Drain Induced Barrier Lowering (DIBL)

Where threshold voltage of MOS decreases as the V_{ds} increases.

channel length reduces $\rightarrow$ potential increases $\rightarrow$ increased depletion layer $\rightarrow$ V_{th}

④ Subthreshold conduction

leakage current flows in $V_{gs} < V_{th}$ due to weak inversion in the channel region causing power dissipation in OFF state transistor.

⑤ Gate leakage

Refers to the condition where, unwanted i^o flow through the gate terminal of MOS due to imperfections in insulating semiconductor oxide layer.

- Due to

- thin gate oxide

- defects & impurities

- Factors influencing

T_{ox} , temp
 V_{gs}

- Impacts

- power dissipation (static)

- Scaling limitations

⑥ Body Effect

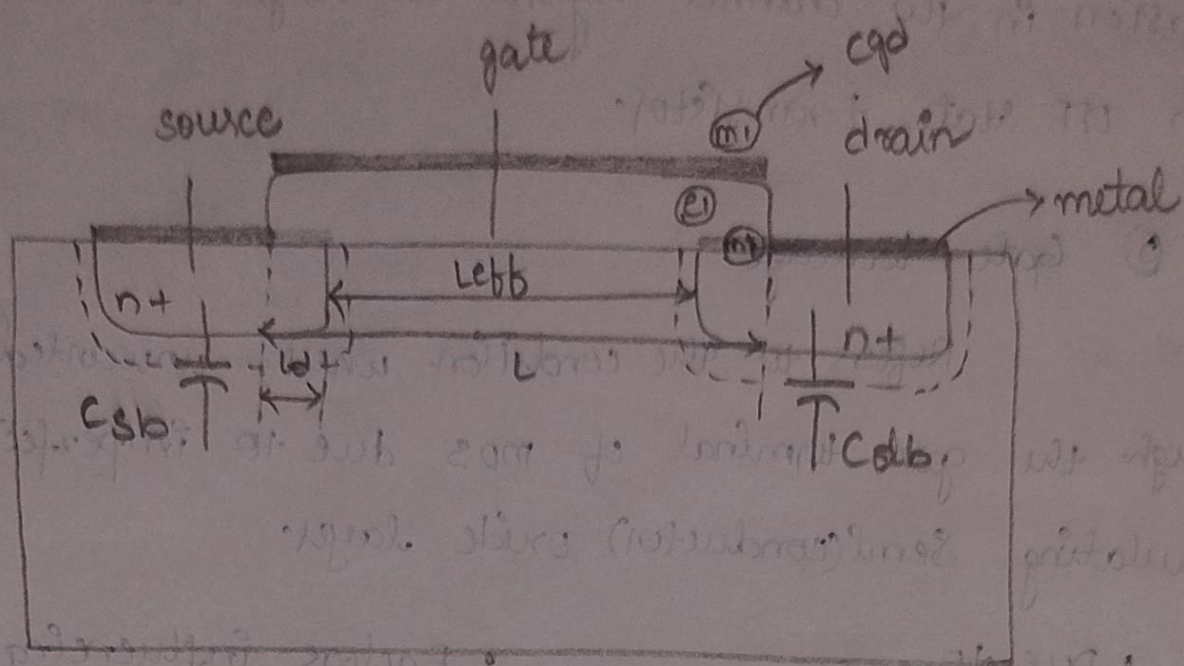
Increasing V_{sb} causes the channel to be depleted of charge carriers & V_{th} is raised.

$$\Delta V_t = \underbrace{\gamma (V_{sb})^{\frac{1}{2}}}_{\rightarrow \text{const.}}$$

$$V_t = V_t(0) + \left(\frac{D}{\epsilon_{ins} \epsilon_0} \right) \sqrt{2 \epsilon_0 \epsilon_{Si} q N} (V_{sb})^{\frac{1}{2}}$$

CV Characteristics

Capacitors in MOSFET



Parasitic Capacitors

$$C_{gb} - C_{gate\ body} = C_{ox} L W$$

$$C_{gs} = C_{ox} L_d W - C_{gate\ source}$$

$$C_{gd} - C_{gate\ drain} = C_{ox} L_d W$$

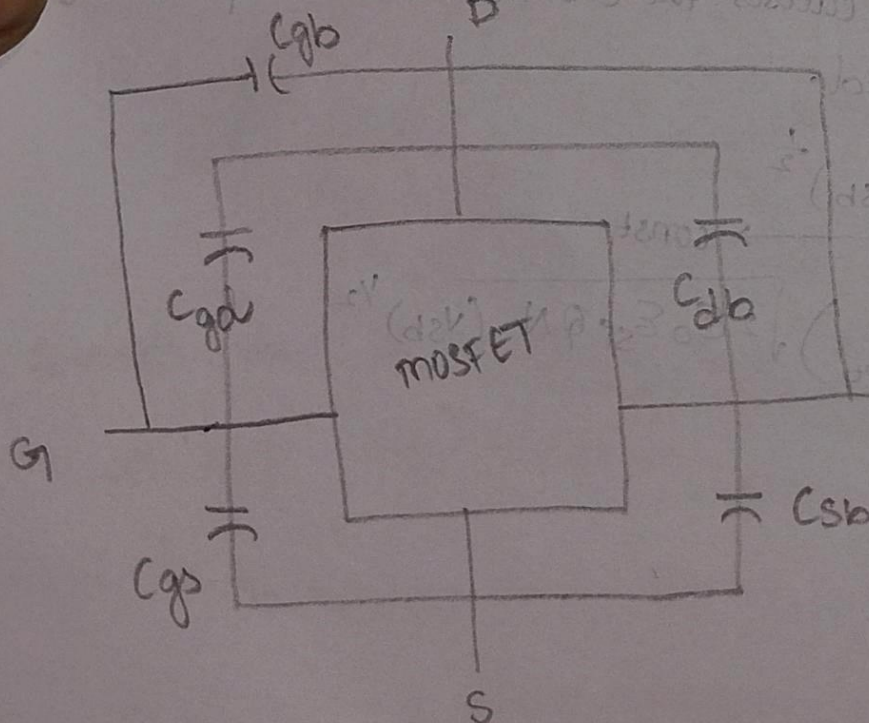
channel capacitance

when some bias is done, diffusion

C_{sb} , C_{db} are formed.

bias - n^+ depletion region

$\rightarrow C_{sb}, C_{db}$



$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

$$C_{gb} = \frac{\epsilon_{ox}}{t_{ox}} L W$$

Gate Capacitance model.

It has two components, the intrinsic (C_{gc}) overlap over the channel and overlap capacitance C_{gal} .

Intrinsic capacitance have 3 components C_{gb} , C_{gs} , C_{gd}

Intrinsic capacitance in different regions

Cutoff - $V_g < V_{th}$, $I_d = 0$

Transistor OFF

channel isn't inverted & charge on gate is matched with opp charge from the body. C_{gb} for $(-ve)$ V_{gs} transistor is in accumulation $\Rightarrow C_{gb} = C_o$ as we increase V_{gs} but less than V_{th} depletion region forms at surface - reducing the capacitance.

Linear - $V_{gs} > V_{th}$

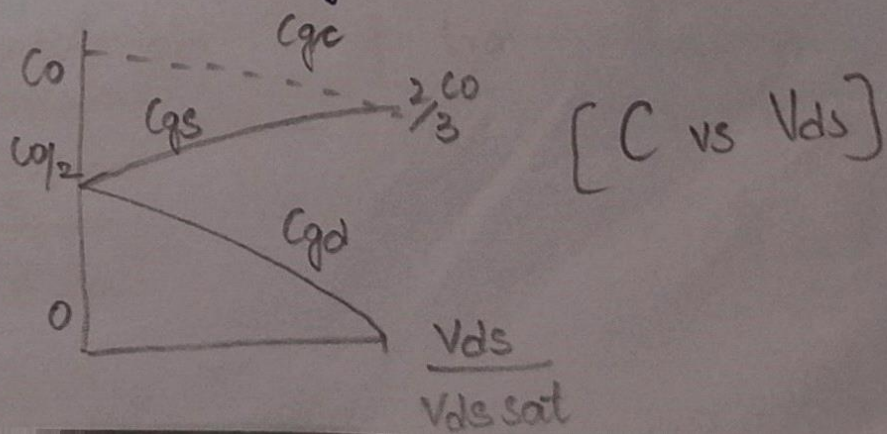
channel is formed so, C_{gb} drops to 0.

at low values of V_{ds} channel charge is roughly shared b/w S & D so, $C_{gs} = C_{gd} = \frac{C_o}{2}$ so we increase V_{ds}

further the depletion layer near drain gets inverted, as so greater fraction of the cap is attributed to source & smaller fraction of drain.

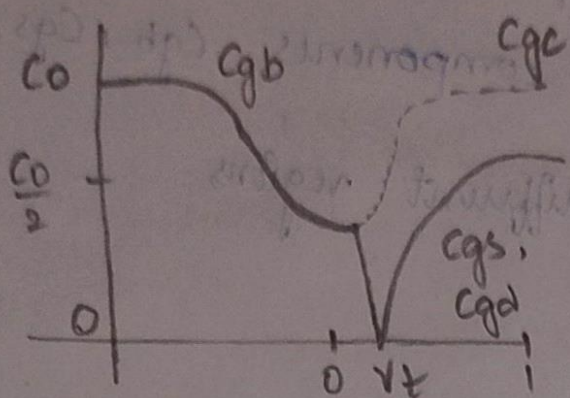
Saturation - $V_{ds} > (V_{gs} - V_{th})$ V_{ds} sat
pinch off
channel will get inverted at drain so,

$$C_{gs} = \frac{2}{3} C_o$$



where,

$$C_{gate} / C_{ge} = C_{gs} + C_{gd} + C_{gb}$$



[C vs Vgs]

Parameters: Cutoff Linear Saturation

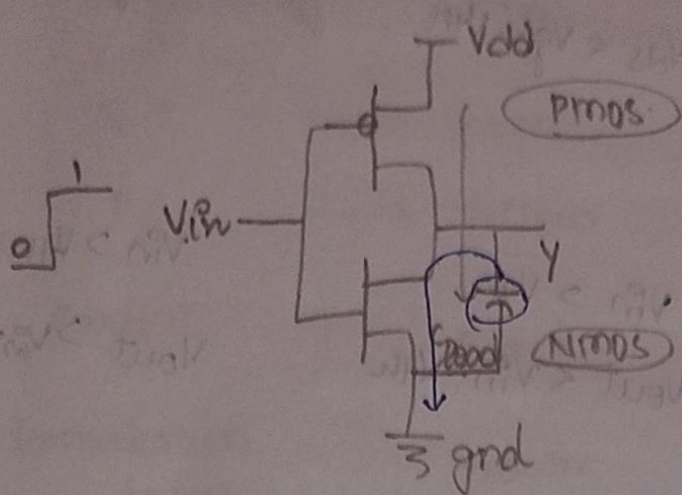
C_{gb}	C_0	0	0
C_{gs}	0	$C_0/2$	$2/3 C_0$
C_{gd}	0	$C_0/2$	0
C_{ge} / C_g	C_0	C_0	$2/3 C_0$

at low values of V_{gs} channel is formed in inversion layer. The depletion of the cap is attributed to source & drain. Further the depletion layer near gates is inverted. As V_{gs} increases the channel depth is negligible and $C_{gs} = C_{gd} = C_0/2$. At $V_{gs} = V_T$, $C_{gs} = C_{gd} = C_0/2$ and $C_{gb} = C_0/2$. At $V_{gs} > V_T$, $C_{gs} = C_{gd} = C_0/2$ and $C_{gb} = C_0/2$. Channel will get inverted to produce inversion.

Transfer Characteristics

CMOS Inverter

[enhancement]

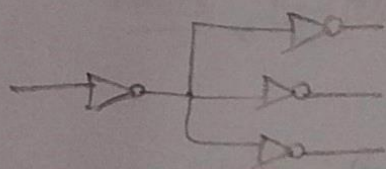


ip 1 op 0 → gnd
ip 0 op 1 → VDD

When $V_{in} = 0$, PMOS is ON [$V_{gs} < V_{th} - ON (-u)$], NMOS is off so y is charging to sound from V_{DD} as C_{load} is charging up so, until $i_p > 0$, $o_p > 1$. When $i_p > 1$, PMOS OFF, NMOS ON, C_{load} have some accumulated voltage which finds a low resistance path to ground given 0 unit $i_p > 1$.

① NO static power consumption.
↓
ideal state

② High impedance → improve fanout

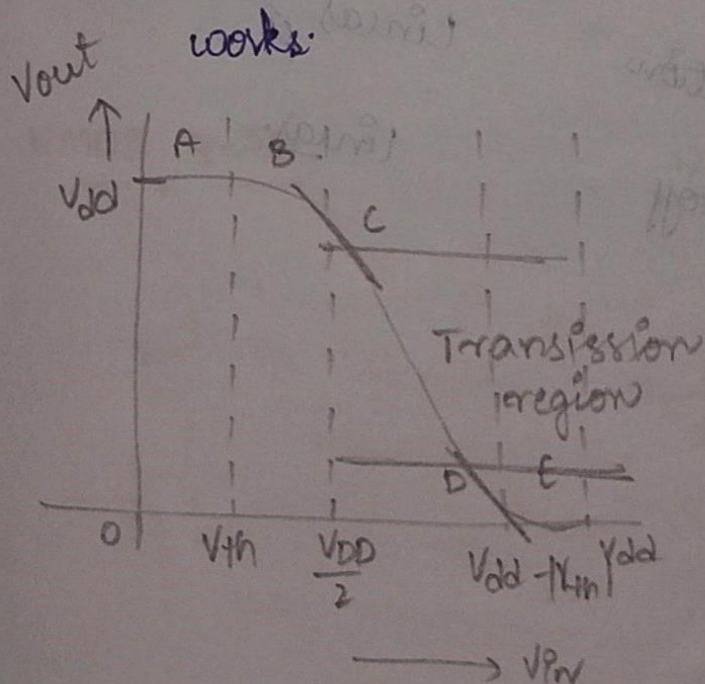


fanout = 3

capability to run n devices

③ High noise margin - SNR is max at op

④ Irrespective of MOS w/l ratio inverter works.



$$\frac{\partial V_{out}}{\partial V_{in}} = -1$$

ip transition happens.

Nmos

Cutoff

Linear

Saturation

$$V_{gs} < V_{th}$$

$$V_{gs} > V_{th}$$

$$V_{gs} > V_{th}$$

$$V_{ds} < V_{gs} - V_{th}$$

$$V_{ds} > V_{gs} - V_{th}$$

Nmos
in
inverter

$$V_{in} < V_{th}$$

$$V_{in} > V_{th}$$

$$V_{in} > V_{th}$$

$$V_{out} < V_{in} - V_{th}$$

$$V_{out} > V_{in} - V_{th}$$

pmos

$$V_{gs} > V_{th}$$

$$V_{gs} < V_{th}$$

$$V_{gs} < V_{th}$$

$$V_{ds} > V_{gs} - V_{th}$$

$$V_{ds} < V_{gs} - V_{th}$$

pmos

in

inverter

$$V_{in} > V_{th} + V_{dd}$$

$$V_{in} < V_{th} + V_{dd}$$

$$V_{in} < V_{th} + V_{dd}$$

$$V_{out} > V_{in} - V_{th}$$

$$V_{out} < V_{in} - V_{th}$$

Region

pmos

Nmos

A

linear

cutoff

B

linear

saturation

C

saturation

Saturation

D

saturation

linear

E

cutoff

linear

STICK x Diagrams

$$(518)4 + 0 = 2$$

metal	Blue	_____
Polysilicon	Red	_____
N +	Green	_____
P +	Yellow	_____
Contact	Black	•
demarkation layer	Brown	-----

$$\text{pmos} - s, d \rightarrow p^+$$

$$b \rightarrow n\text{type}$$

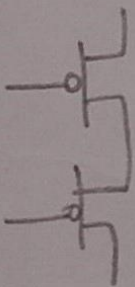
$$\text{NMOS} - s, d \rightarrow n^+$$

$$b \rightarrow p\text{type}$$

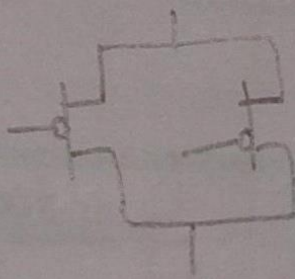
NMOS with depletion $\rightarrow$ Red over green with yellow box at center.

- PMOS — parallel
- NMOS — series
- + PMOS — series
- + NMOS — parallel

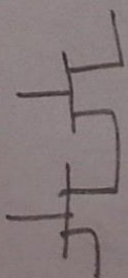
PMOS x Series



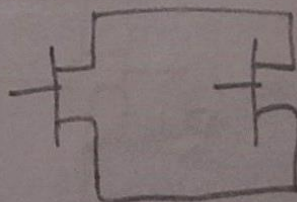
Parallel - PMOS



NMOS series

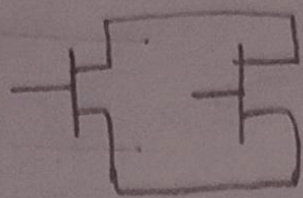
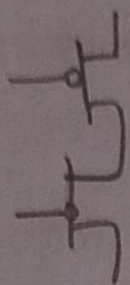


NMOS x parallel

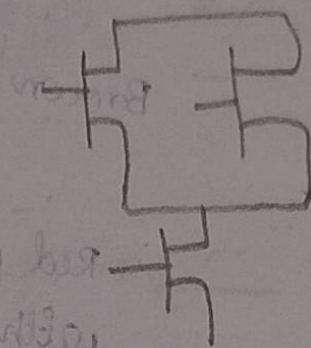
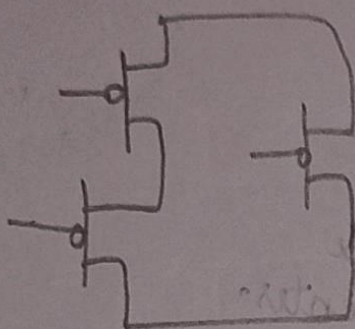


$$Y = \overline{D + A(B+C)}$$

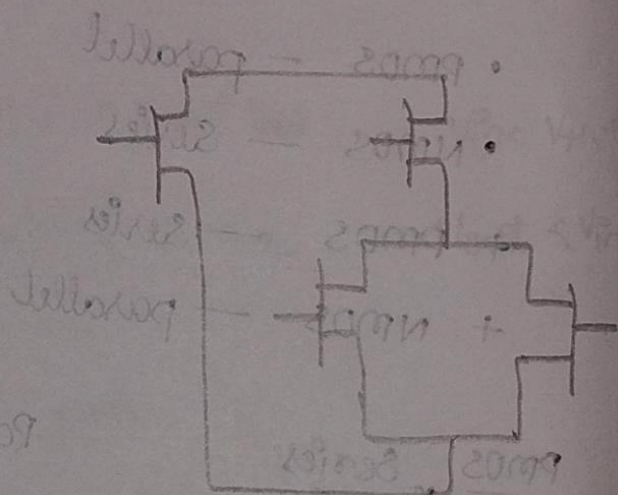
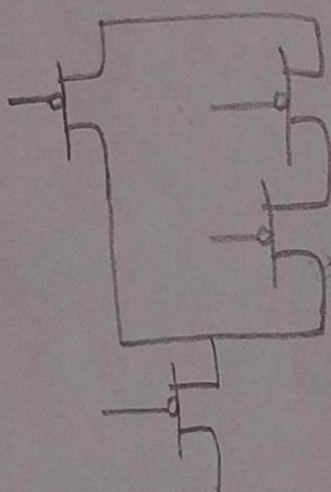
$$B+C$$



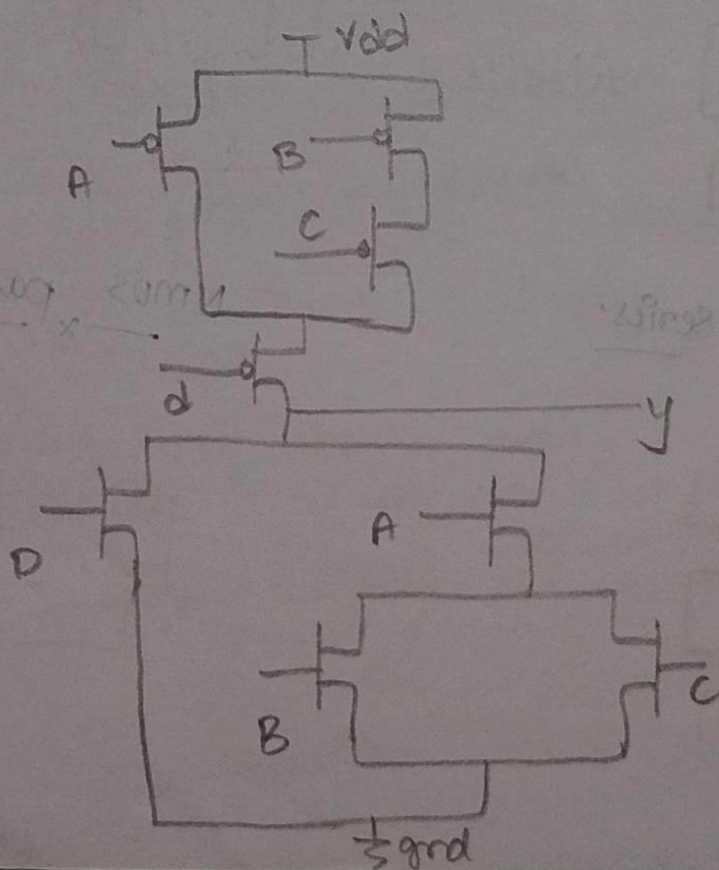
$$A(B+C)$$



$$D + A(B+C)$$

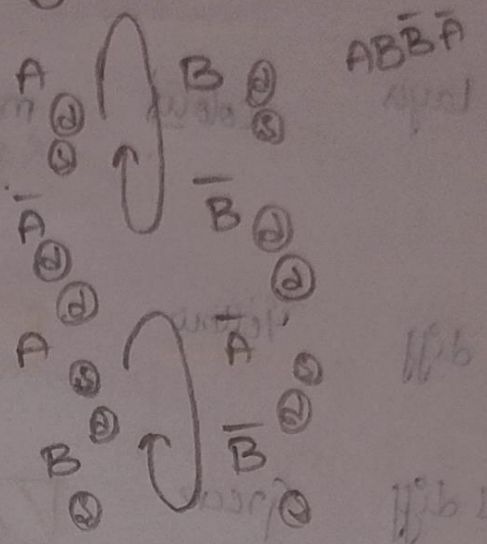
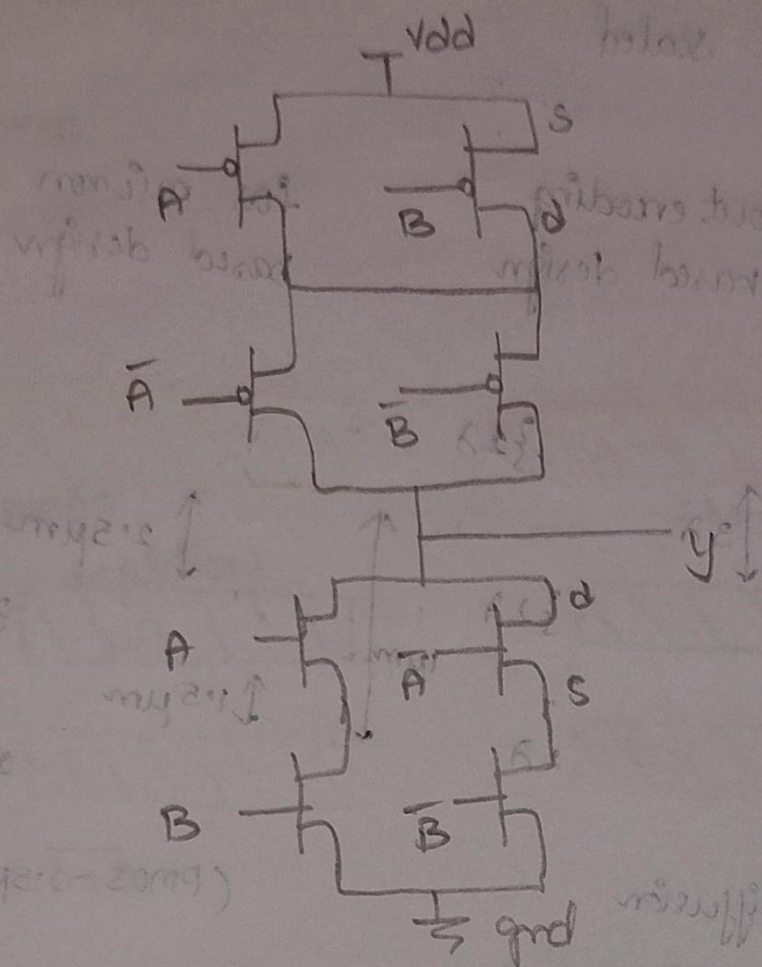


$$Y = \overline{D + A(B+C)}$$



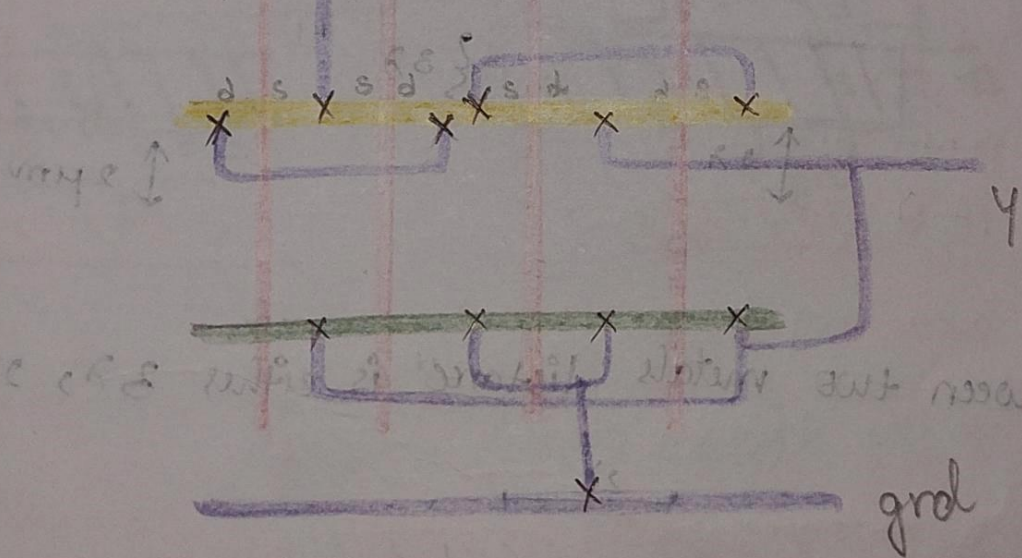
$$Y = \overline{AB} + A\overline{B} \Rightarrow \overline{AB} + A\overline{B} = (A+B)(\overline{A}+\overline{B})$$

$$= \overline{AB + \overline{A}\overline{B}}$$



Between a buffer and a multiplexer, a multiplexer is needed.

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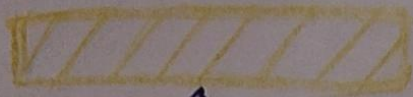
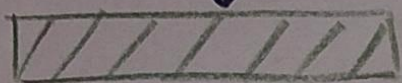
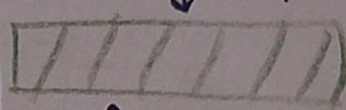
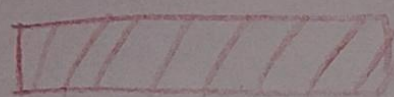
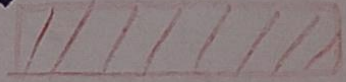
Between a buffer and a multiplexer, a multiplexer is needed.

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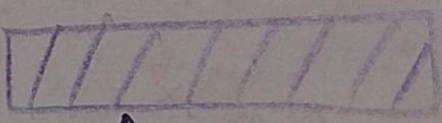
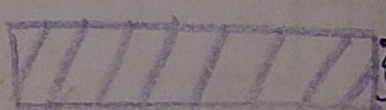
Between a buffer and a multiplexer, a multiplexer is needed.

Layouts

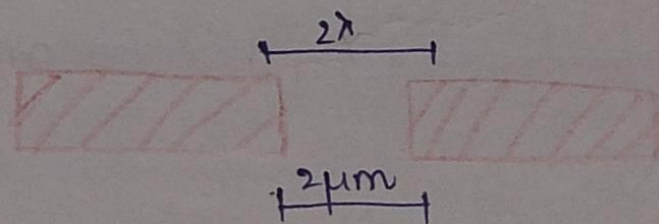
micron - layers are fixed
 λ - layers can be scaled

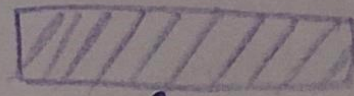
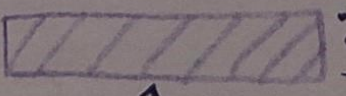
Layers	colour	mask layout encoding for λ based design	for micron based design
P diff	Yellow	 $\{ 2\lambda$	 $2\mu m$
N diff	Green	 $\{ 2\lambda$	 $3\mu m$
Poly	Red	 2λ any diffusion	 $2\mu m$ (Nmos) (Pmos - $2.5\mu m$)

between 2 polysilicons of pmos, poly = $3\mu m$
 " " " " nmos, poly = $2\mu m$

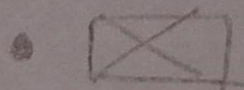
Metal 1,2	Blue	 $\{ 3\lambda$	 $\{ 2.5\mu m$
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Between two metals distance is either 3λ , $2.5\mu m$

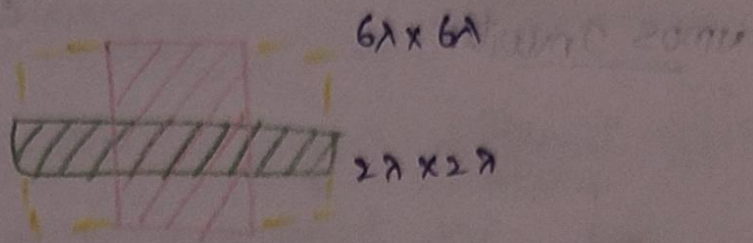


Metal 3,4		 $\{ 4\lambda$	 $\{ 3\mu m$
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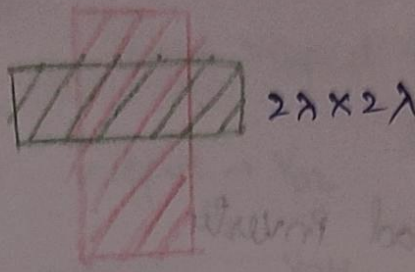
Contact Black



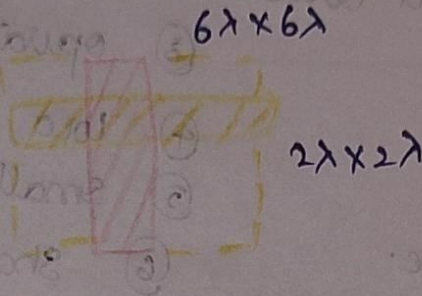
NMOS depletion



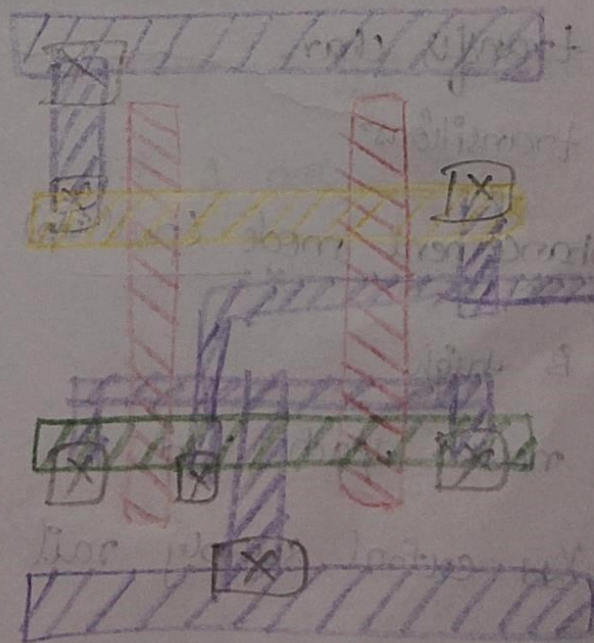
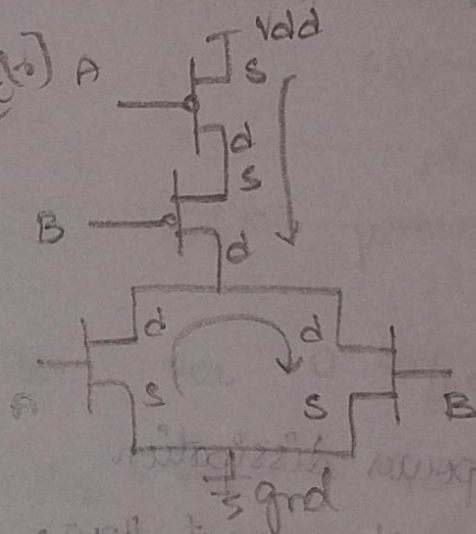
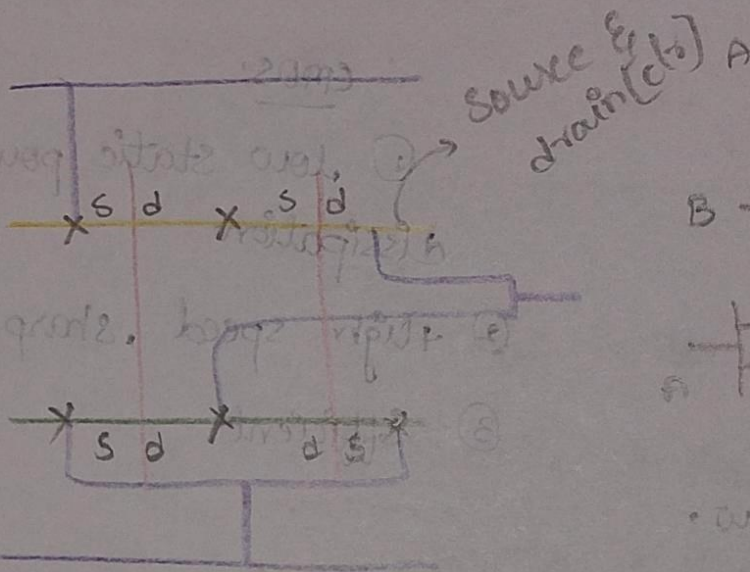
NMOS enhancement



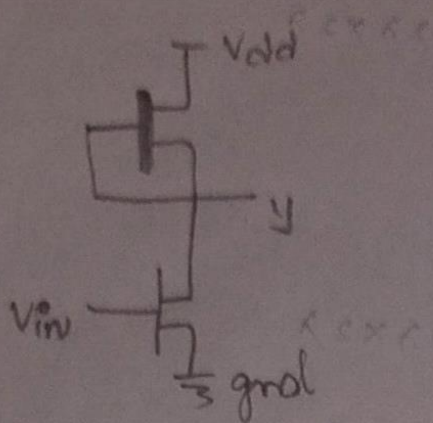
PMOS depletion



$$Y = \overline{A+B}$$



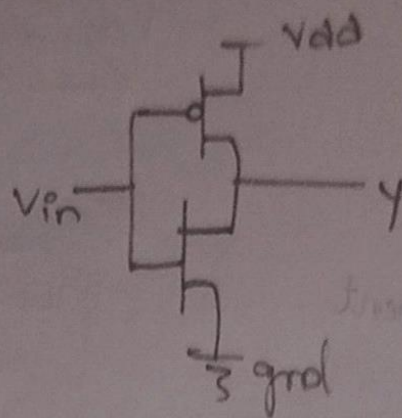
NMOS Inverter



depletion load Inverter

- ① Gate is connected source
it is always on.

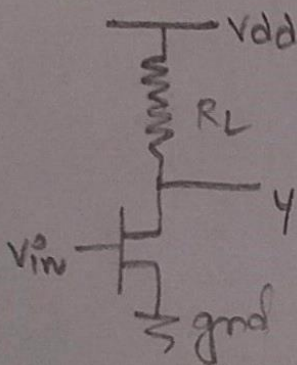
CMOS Inverter



- ② Better noise margin
- ③ operates in linear / sat
- ④ low power dissipation at '0'
- ⑤ small area required mainly
- ⑥ Sharp VTC

Load Resistance

Resistive load Inverter



- ① High power dissipation
because of const current flow.
- ② Large area req for R
- ③ poor voltage transfer char
due to non-sharp transitions?

CMOS

- ① low static power dissipation
- ② High speed, sharp VTC
- ③ Efficient

Preference

CMOS >>> dep > enhan > Resis

NMOS Enhancement mode (pull up)

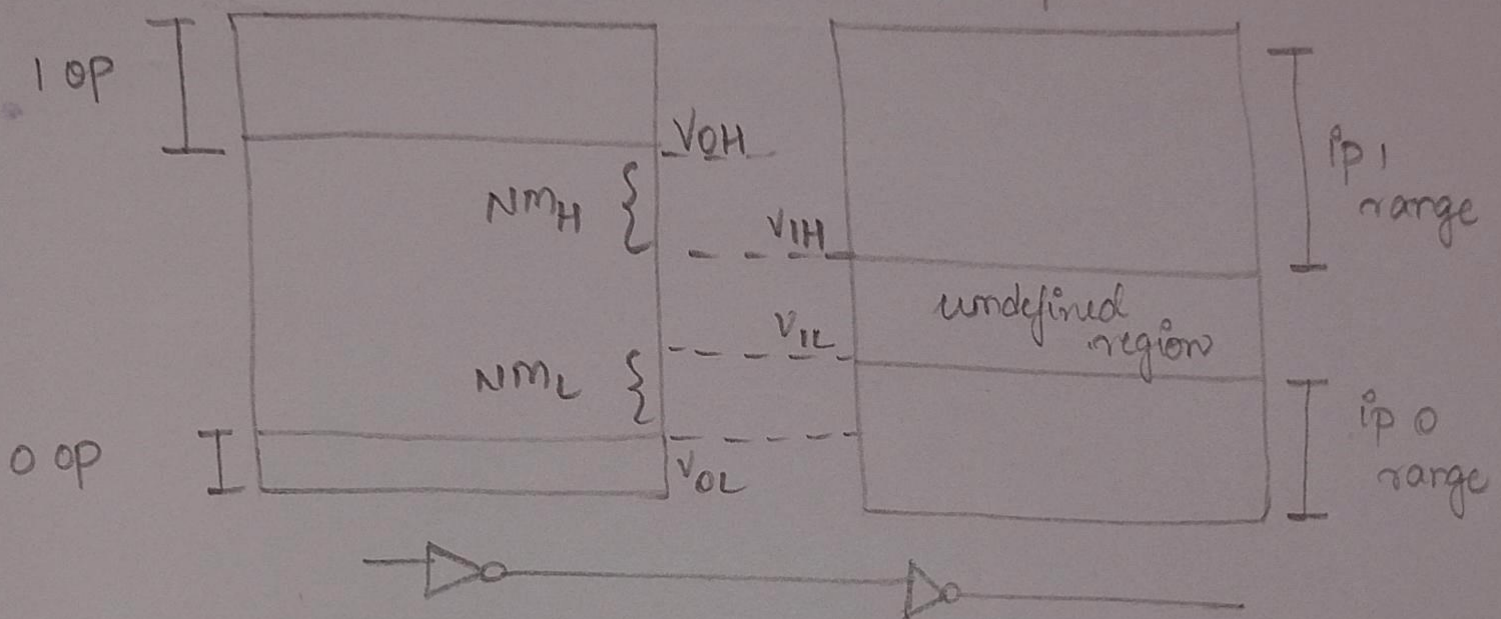
- ① Dissipation is high
- ② V_{out} can never reach V_{dd}
- ③ if $V_g > V_{dd}$ external supply rail req.

NOISE MARGIN.

Amt of noise inputs can reject without affecting the output.

$$\begin{aligned}
 & \text{noise margin high/low} \leftarrow \begin{cases} NML = V_{IL} - V_{OL} & \text{input low, output low} \\ NMH = V_{OH} - V_{IH} & \text{input high, output high} \end{cases}
 \end{aligned}$$

when
input $\rightarrow V_{IH}$ output $\rightarrow V_{OL}$
 $\downarrow$ $\downarrow$
 V_{IL} V_{OH}
OP IP



WKT,

$$V_{OH} = V_{DD}, V_{OL} = 0$$

$$\Rightarrow NM_L = V_{IL} - V_{OL} = V_{IL}$$

$$NM_H = V_{OH} - V_{IH} = V_{DD} - V_{IH}$$

when I assume

$$V_{IL} = V_{IH} = \frac{V_{DD}}{2} \text{ I can say the probability}$$

of noise of both 0 & 1 are rejected.